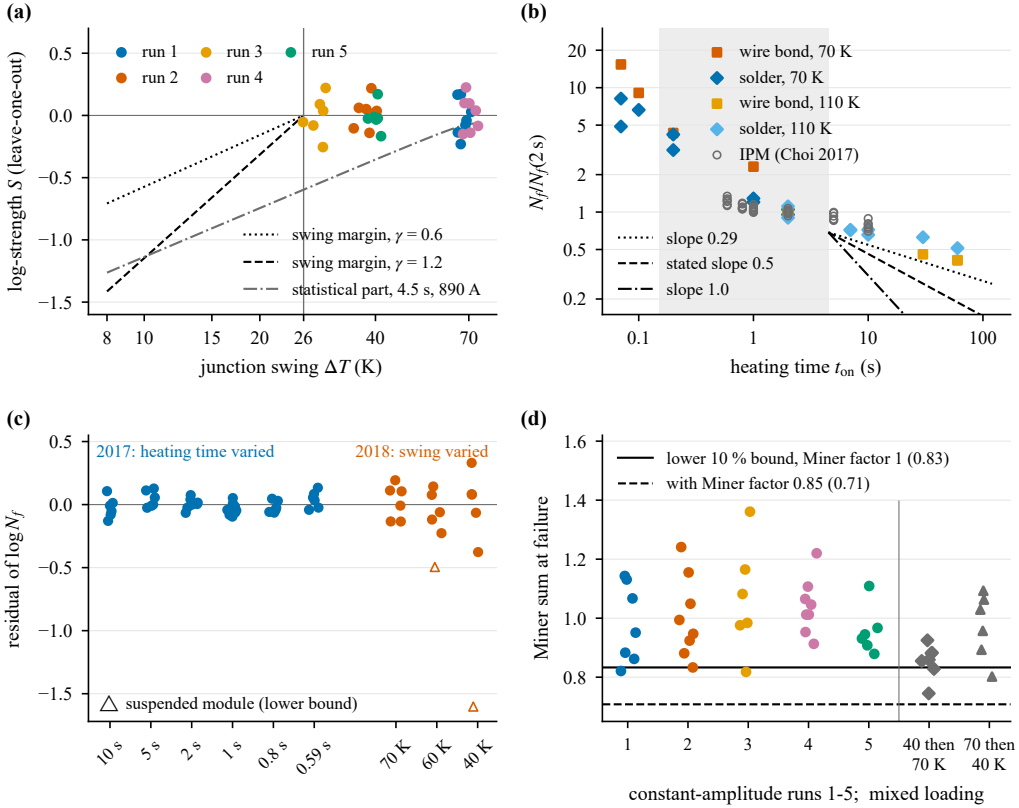

(a)
run 1
run 2
run 3
run 4
run 5
swing margin, γ = 0.6
swing margin, γ = 1.2
statistical part, 4.5 s, 890 A
log-strength S (leave-one-out)
junction swing ΔT (K)
(b)
wire bond, 70 K
solder, 70 K
wire bond, 110 K
solder, 110 K
IPM (Choi 2017)
slope 0.29
stated slope 0.5
slope 1.0
N_f/N_f(2 s)
heating time t_on (s)
(c)
2017: heating time varied
2018: swing varied
suspended module (lower bound)
residual of log N_f
10 s
5 s
2 s
1 s
0.8 s
0.59 s
70 K
60 K
40 K
(d)
lower 10 % bound, Miner factor 1 (0.83)
with Miner factor 0.85 (0.71)
Miner sum at failure
40 then 70 K
70 then 40 K
constant-amplitude runs 1-5; mixed loading